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CE205
Databases and Information Retrieval

Types of FDs, 2NF, 3NF, BCNF

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Most of the slides based on

- *Database Systems: A Practical Approach to Design, Implementation, and Management* by Conoly and Begg, Global Edition/6th Edition.
- *Database Systems: Introduction to Databases to Data Warehouses* by Jukic, Vrbsky, Nestorov, sharma, 2nd Edition.

Learning Objectives

- Graphical representation of FDs
- What are Partial FD Transitive FD, and Full Key FD
- Dependency Preserving Decomposition
- Lossless Join Decomposition
- How to Normalize a relation in 2NF, 3NF and BCNF

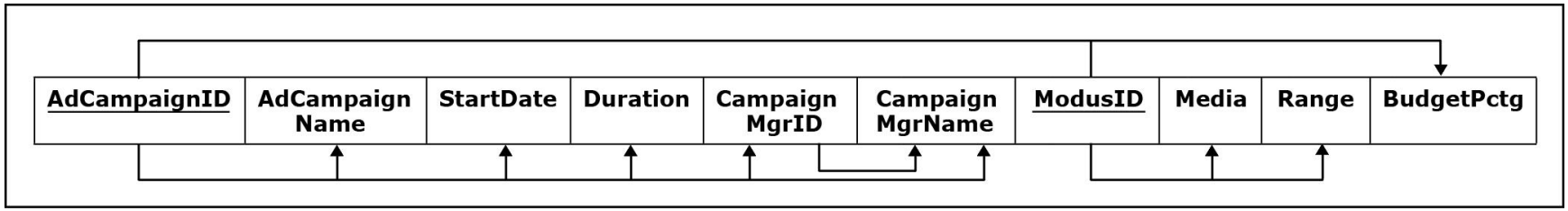
Graphical Representation of FDs of Relation AD COMPAIGN MIX

(Set 1) AdCampaignID → AdCampaignName, StartDate, Duration, CampaignMgrID, CampaignMgrName

(Set 3) CampaignMgrID → CampaignMgrName

(Set 4) ModusID → Media, Range

(Set 5) AdCampaignID, ModusID → BudgetPctg



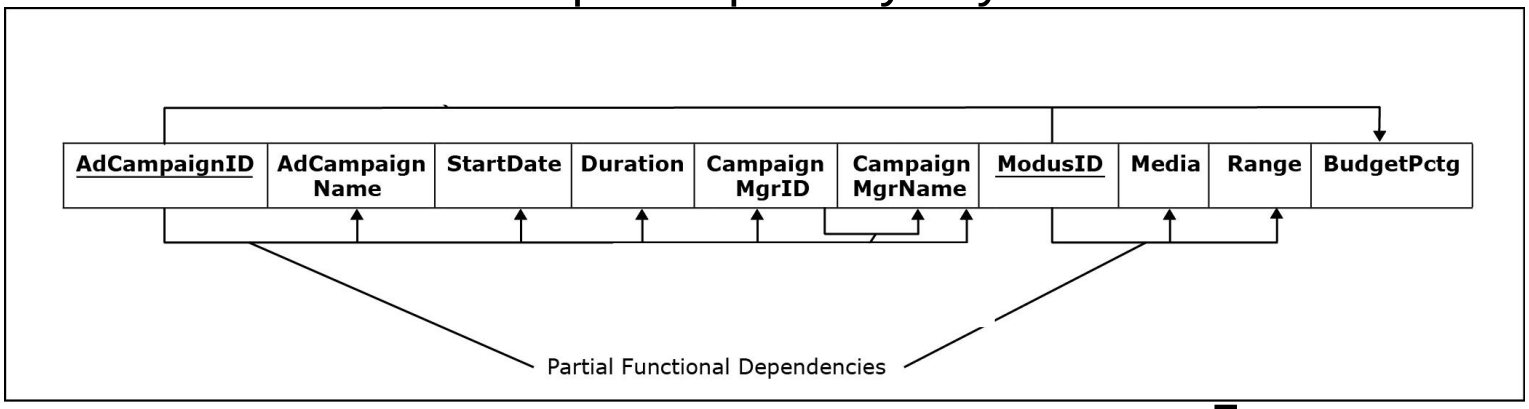
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Types of Functional Dependencies

- The functional dependencies that are used as a basis for the typical normalization process can be classified in one of the three categories:
  - Partial functional dependency
  - Full key functional dependency
  - Transitive functional dependency

Partial Functional Dependency

- Partial functional dependency - occurs when a proper subset of the primary key in a relation determines non-prime attribute(s)
  - Only composite primary keys can have proper subsets, while single-column primary keys do not have proper subsets.
  - Hence, partial functional dependency can occur only in cases when a relation has a composite primary key

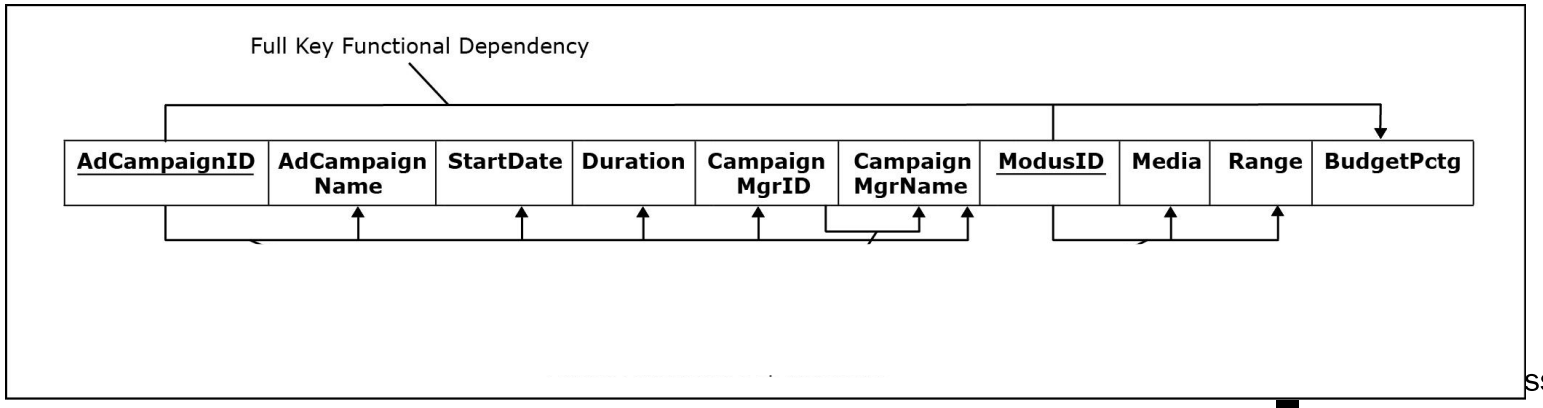


Primary Key= {AdCampaignID, ModusID}

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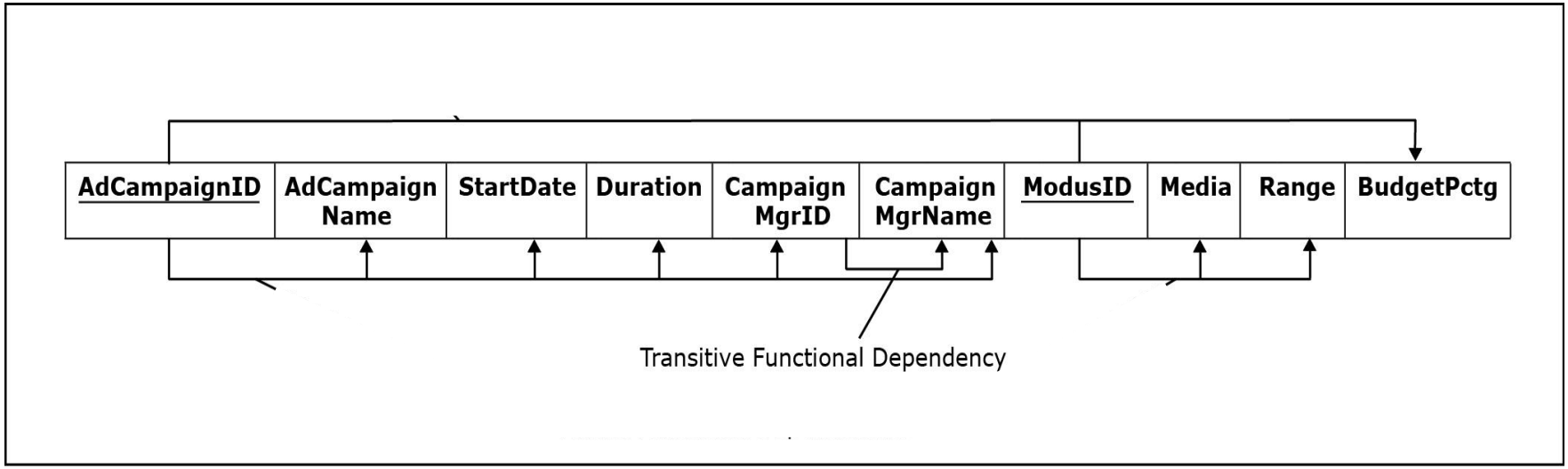
Full Key Functional Dependency

- Full key functional dependency - occurs when a primary key functionally determines the column of a relation and no separate component of the primary key partially determines the same column
  - If a relation has a single component (non-composite) primary key, the primary key fully functionally determines all the other columns of a relation



Transitive Functional Dependency

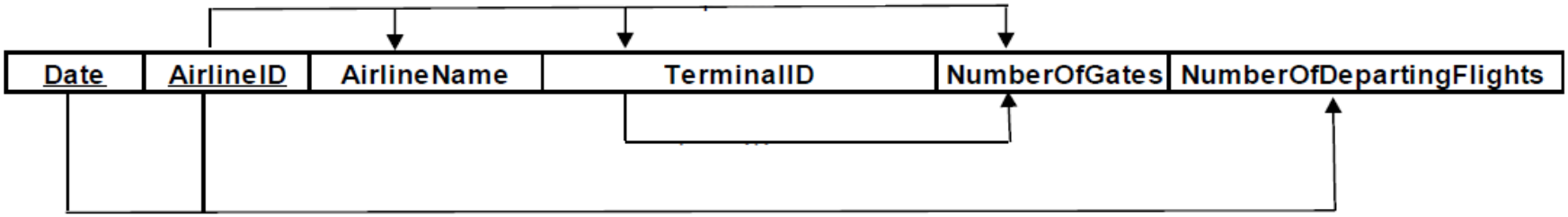
- Transitive functional dependency - occurs when non-prime attribute(s) functionally determine other non-prime attribute(s) of a relation



Identify Partial, Full Key and Transitive FDs in Relation AIRPORT

- AirlineID → AirlineName, TerminalID, NumberOfGates
- TerminalID → NumberOfGates
- {Date, AirlineID} → NumberOfDepatingFlights

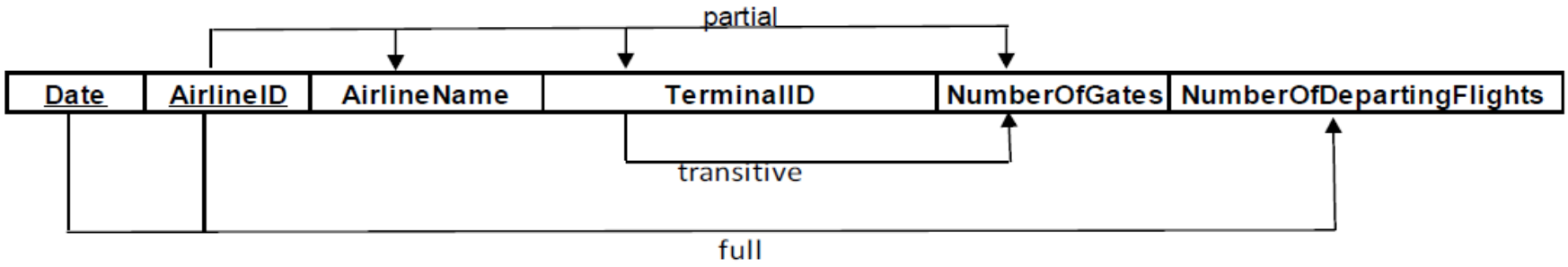
- Graphical representation of FDs



Primary Key= {Date, AirlineID}

Identify Partial, Full Key and Transitive FDs in relation Airport--Solution

- AirlineID → AirlineName, TerminalID, NumberOfGates
- TerminalID → NumberOfGates
- {Date, AirlineID} → NumberOfDepatingFlights



Primary Key= {Date, AirlineID}